

AI Swarm Drones Find Survivors **3-4x Faster** Than Traditional SAR

SYSTEM PERFORMANCE TELEMETRY

6X6
VERIFIED

SUBSYSTEM MOAT ARCHITECTURE

3 CORE ENGINEERING
MOATS

DETECTION RATE

**78-
85%**

First-pass
probability

AREA SPEED

**10-
18
min**

Per sq mile
(vs 2-3
hrs)



Subsystem A Moat: PX4 + VIO EKF2 Autopilot

50Hz
OFFBOARD

Decentralized 3D ORCA anti-collision with zero tower reliance and active GPS-denied flocking algorithms.

COMMS PROTOCOL

**Mesh
Net**

GPS-Denied
Autonomous

SUSTAINED AUTONOMY

**24/7
(11+
Days)**

Continuous
field
sorties



Subsystem B Moat: Deep JSCC Neural Video

-5dB
RESILIENT

Zero-cliff neural semantic video encoding with graceful degradation under severe mountain ravine occlusion.

FALSE POSITIVES

12%

Down from
35% legacy

DEPLOYMENT COST

\$12K

~~\$15K~~

Per sortie
deployment



Subsystem C/D Moat: 3D DEM Terrain Raycasting

3.59cm
ACCURACY

Sub-15ms edge YOLOv8-Pose with WGS84 elevation raycasting, eliminating 15-30m flat-Earth geolocation errors.

TAM
PROJECTION

\$8.94B
Market
2033

DEMAND
DELTA

+412%
Surge

REGULATORY

37
Nations
Mandate

FEMA
ALLOCATION

\$180M
in
2025